

Hope Artificial Intelligence



Assignment-Regression Algorithm

Download Set: •

Google Drive Link [Click here](#) .

Git Hub Link: https://raw.githubusercontent.com/RamishaRaniK/dataset/main/insurance_pre.csv

Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

As a data scientist, you must develop a model which will predict the insurance charges.

1.) Identify your problem statement

The objective is to develop an AI model that predicts the insurance charges of the customer based on several parameters provided in the input data such as (age, sex, has the person have the smoking habit or not, how many children they have). Using the provided dataset, we have to develop a best model using various algorithms and finalize the best model that predicts accurate insurance charges for the client.

2.) Tell basic info about the dataset (Total number of rows, columns)

Total Number of Rows in the dataset: 1339

Total Number of Columns in the dataset: 6

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

This dataset has categorical values for multiple attributes. The categorical input data can't be handled by python. So, as a pre-processing step, turned the categorical data into nominal data (One Hot Encoding) using **get_dummies** function.

```
: dataset=pd.get_dummies(dataset, dtype=int, drop_first=True)
```

- 4.) Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

After creating and trying with various model, I have finalized the best model using Random Forest algorithm with the r2_score value of **0.871**

- 5.) All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

Refer: <https://github.com/akshay811998/Regression-Assignment/blob/main/R-Score%20Values%20Baby%20Steps%203.pdf>

- 6.) Mention your final model, justify why u have chosen the same.

Finalized the model using random forest and the reason why I've chosen random forest among Decision Tree, MLR and SVM is because, MLR and SVM provided less r2_score comparative to Decision Tree and Random Forest. The input given is large dataset which has 1000+ rows. So while testing with decision tree and random forest, random forest provided reliable and faster results when decision tree took a lot of time to generate output and as well as r2_score value is better in random forest

Refer: <https://github.com/akshay811998/Regression-Assignment/blob/main/Insurance Charges Prediction RF.ipynb>

Refer: <https://github.com/akshay811998/Regression-Assignment/blob/main/Insurance Charges Prediction RF Deploy.ipynb>

Refer: <https://github.com/akshay811998/Regression-Assignment/blob/main/ICP finalized model randomforest.sav>

Kindly create Repository in the name Regression Assignment.

Refer: <https://github.com/akshay811998/Regression-Assignment>

Upload all the ipynb and final document in the pdf

Refer: <https://github.com/akshay811998/Regression-Assignment>

Communication is important (How you are representing the document.)